

n8n Workbook

Build your first AI-powered customer support workflow — step by step.

A hands-on guide for beginners. By the end, you'll have a working automation that reads customer emails, classifies them with AI, drafts replies, and logs everything to a sheet your team can actually use.

SKILL LEVEL

Beginner

TIME

45–60 min

YOU'LL USE

n8n • Gmail • OpenAI • Sheets

CONTENTS

What's inside

Eight short parts. Skip around as needed, but if this is your first time, read in order.

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• HOW TO USE THIS WORKBOOK

Each step is small on purpose. Don't skip the test runs — they're how you catch mistakes early. If a step doesn't work, jump to the troubleshooting section. Every problem you'll hit is in there.

PART ONE

Welcome & what you'll build

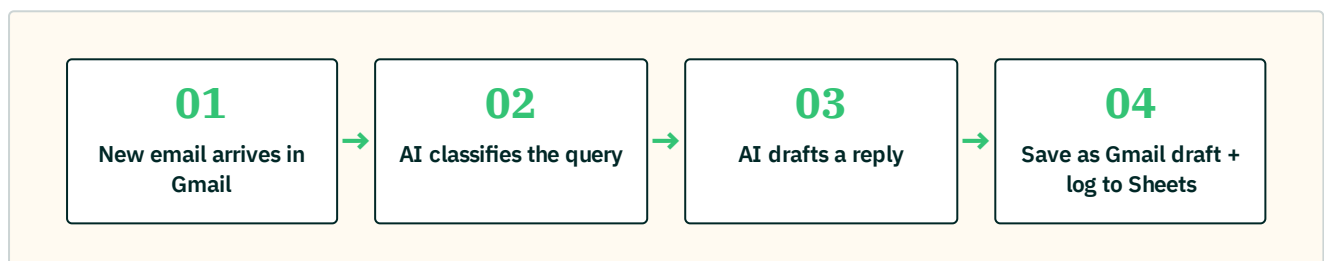
A quick tour of the workflow you're about to ship, what you need before you start, and what you'll walk away knowing.

The workflow you're going to build

By the end of this workbook you'll have a real, working automation. Every time a new email lands in your Gmail inbox, this workflow will read it, figure out what the customer wants, draft a reply for your team to review, and log a row to a Google Sheet so nothing slips through the cracks.

This isn't a toy example. The same pattern runs in thousands of small support teams today. You're not learning n8n in the abstract — you're shipping something useful.

What it does, end to end



What you'll learn along the way

- How a workflow comes together — triggers, nodes, and how data flows between them
- How to connect Gmail and Google Sheets safely using OAuth
- How to use OpenAI inside n8n for both classification and writing
- How to test a workflow piece by piece (and why this matters more than building fast)
- The "human-in-the-loop" principle — why we draft replies instead of auto-sending them

• A NOTE ON SAFETY

This workflow saves replies as **drafts**, not sent emails. That's deliberate. AI can sometimes misread context or make confident-sounding mistakes. A 5-second human review keeps your customer experience sharp and your reputation intact. We'll come back to this idea throughout.

Before you start

You don't need to be a developer to follow this workbook. You do need a few accounts ready to go. Spend 10 minutes ticking these off before you open n8n — it'll save you switching contexts later.

Your prerequisites checklist

- ☐ A working Gmail account (personal or workspace, both fine)
- ☐ An OpenAI account at `platform.openai.com` with API access turned on
- ☐ About **\$5 in OpenAI credits** — you'll spend pennies, but the API needs a billing method on file
- ☐ Access to Google Sheets (free; same Google account as your Gmail is easiest)
- ☐ A modern browser — Chrome, Edge, Firefox, or Safari, latest version
- ☐ Roughly 45–60 minutes of focused time

How much will this cost?

n8n Cloud	Free during the 14-day trial. Plans start at \$24/month after, but you can also self-host for free later.
OpenAI usage	Roughly \$0.001 to \$0.005 per email with the <code>gpt-4o-mini</code> model. Processing 1,000 emails costs about \$2–\$5.
Gmail & Google Sheets	Free with any Google account.

• HEADS UP ON API KEYS

You'll generate an OpenAI API key in the next part. Treat it like a password: don't paste it into chats, screenshots, or shared docs. If it leaks, anyone can spend money on your account. n8n stores it securely once you save it — you only handle it once.

What this workbook assumes you already saw in the workshop

This guide is a companion, not a replacement. We assume you've watched (or been walked through) the basics of opening n8n, dragging in a node, and clicking around the canvas. If any of that feels foggy, the next part is a quick refresher.

PART TWO

A quick n8n refresher

If you remember what nodes, triggers, and executions are, skim this. If they sound vague, slow down — these five concepts are everything.

The five ideas that run all of n8n

n8n looks like a complicated tool the first time you see it. It isn't. The whole thing is built around five ideas. Once these click, every node and every workflow makes sense.

1. Workflow

A workflow is a chain of steps that runs from left to right. You build it on a visual canvas by dragging nodes in and connecting them with lines. The customer support automation you're about to build *is* one workflow.

2. Node

A node is one step. "Send a Slack message," "Read a row from Google Sheets," "Call OpenAI" — each of these is a node. n8n ships with 400+ pre-built nodes for popular tools, plus a generic HTTP node for anything else.

3. Trigger

A trigger is a special node that *starts* a workflow. Without a trigger, nothing happens. The most common triggers are: a webhook, a schedule (every X minutes), or an event from a connected app (like "new email in Gmail" — the one you'll use today).

4. Execution

An execution is a single run of your workflow. Each time a new email arrives, that's one execution. n8n keeps a log of every execution so you can see what worked, what failed, and what data passed through each node.

5. Credential

A credential is your saved authentication for a service — your Gmail OAuth token, your OpenAI API key, etc. You set them up once, and any node can reuse them. Credentials are encrypted and never shown back to you in plain text after you save them.

• THE MENTAL MODEL

Think of n8n as a kitchen. **Workflows** are recipes. **Nodes** are the steps in a recipe. **Triggers** are the timer or doorbell that tells you to start cooking. **Executions** are the meals you actually serve. **Credentials** are the keys to the pantry.

The interface, in 60 seconds

When you open a workflow, you'll see roughly the same layout every time. Here's what you're looking at.



Screenshot: n8n workflow editor

Drop a screenshot of the empty workflow canvas here, with the canvas, sidebar, and top bar visible.

THE CANVAS

The big space in the middle. This is where your workflow lives. You'll drag nodes onto it and connect them with lines. Click and drag to pan. Scroll to zoom.

THE "+" BUTTON (OR "ADD FIRST STEP")

This is your entry point to adding nodes. Click it and a search panel opens — type the name of any service to find its node.

THE TOP BAR

Three things you'll touch often:

- **The workflow name** on the left — click to rename
- **The Active / Inactive toggle** on the right — flips your workflow on or off in production
- **The Save button** — n8n auto-saves, but a manual save (Cmd/Ctrl + S) never hurts

THE RIGHT-HAND PANEL (WHEN YOU CLICK A NODE)

This is where you configure each node. It has three sections: **Parameters** (what the node does), **Settings** (advanced options, retries, etc.), and **Input/Output** (the data going in and the data coming out — your debugging best friend).

EXECUTIONS TAB

Up at the top, switch from "Editor" to "Executions" to see every past run of your workflow. Failed executions show in red. Click any one to see exactly what happened, node by node.

- **TWO SHORTCUTS THAT CHANGE EVERYTHING**

Drag data, don't type it. When you're configuring a node, the left side of the panel shows the data from previous nodes. Drag fields directly into your config — n8n writes the expression for you.

Test as you go. Every node has an "Execute step" button. Click it after every change. You'll catch problems instantly instead of two hours later.

PART THREE

Setup — before you build

Three accounts, ten minutes. Get these out of the way and the build itself will fly.

Three things to do first

Sign up for n8n Cloud, grab an OpenAI API key, and create a fresh Google Sheet for logging. Once those are done, you can stay inside n8n for the rest of the workbook.

01 Sign up for n8n Cloud

n8n Cloud is the hosted version. No servers, no Docker, no installs — just a URL and a login. The free 14-day trial gives you everything you need for this workbook.

1. Open `https://app.n8n.cloud` in your browser
2. Click **Get started for free**
3. Enter your work email and a strong password
4. Verify your email (check inbox; the link expires after 24 hours)
5. Pick a subdomain — something like `yourname.app.n8n.cloud`. This is your private n8n URL forever; pick something you'll remember.
6. Skip the onboarding tour for now (or do it — your call)



Screenshot: n8n Cloud dashboard, freshly signed up

Drop a screenshot of the empty dashboard here.

• IF YOU'D RATHER SELF-HOST

n8n is open-source and free if you run it yourself with Docker. We're using Cloud here because it's the fastest path for beginners. Once you're comfortable, switching to self-hosted is a 10-minute job — nothing in this workbook will need to change.

02 Get your OpenAI API key

An API key is what lets your workflow talk to OpenAI. Generating one takes two minutes.

1. Open `https://platform.openai.com` and sign up (or log in)
2. In the left sidebar, click **API keys**
3. Click **+ Create new secret key**
4. Name it `n8n workshop` so you remember what it's for
5. Click **Create secret key**
6. Copy the key (it starts with `sk-`) and paste it somewhere safe — a password manager is best

- **YOU ONLY SEE THE KEY ONCE**

OpenAI shows the full key exactly one time. After you close the dialog, you can never see it again. If you lose it, you'll have to delete the key and make a new one — no big deal, just a hassle.

ADD BILLING (REQUIRED)

7. In the left sidebar, click **Billing**
8. Click **Add payment method** and enter a card
9. Add a starting credit balance — **\$5 is plenty** for this workbook and weeks of testing afterwards
10. Optionally, set a usage limit so you can never accidentally spend more than you intended

- **WHY THIS IS SO CHEAP**

You'll use the `gpt-4o-mini` model — fast, accurate, and currently priced at fractions of a cent per token processed. Even at heavy usage, your bill should stay under a coffee per month. We'll point at the exact model in Step 7.

03 Create your logging sheet

Every customer query that comes in gets logged to a Google Sheet — one row per email. This becomes your team's living tracker. You'll spend two minutes setting it up now and never have to think about it again.

1. Open `https://sheets.google.com` and click the big **+ Blank** tile
2. Click the title at the top-left and rename it to **Customer Support Log**
3. In **row 1**, type these column headers, one per cell:

Timestamp	From	Subject	Category	Priority	Summary	Draft Status
-----------	------	---------	----------	----------	---------	--------------

4. Bold row 1 so the headers stand out (Cmd/Ctrl + B)
5. That's it. Don't add formatting, formulas, or filters yet — n8n needs the sheet plain to write to it cleanly.



Screenshot: Google Sheet with seven column headers

Drop a screenshot of the empty sheet with headers here.

- **WHY EXACT COLUMN NAMES MATTER**

n8n maps your data to columns by header name. If your headers say `Subject` but your workflow tries to write to `subject`, the data goes nowhere. Type the headers exactly as written above — including capitalization.

You're set up. Take a breath.

Three accounts, three keys, one sheet. The hard part is over. From here, everything happens inside n8n — and you'll be done in about 30 minutes.

PART FOUR

Build your workflow

Ten focused steps. Test after each one. By the last step, you'll have a working AI-powered support automation running on your account.

Start with an empty canvas

04 Create a new workflow

1. From your n8n dashboard, click **+ Create Workflow** (top-right)
2. You'll land on an empty canvas with a single placeholder that says *"Add first step..."*
3. At the very top of the page, click on the default workflow name (it'll say something like `My workflow 1`) and rename it to:

Customer Support Triage

4. Hit Enter to confirm. The name now shows in the top bar.
5. Press **Cmd/Ctrl + S** to save.



Screenshot: Empty canvas with renamed workflow

Drop a screenshot showing the renamed "Customer Support Triage" workflow with empty canvas.

- **GET INTO THE HABIT EARLY**

n8n auto-saves, but pressing **Cmd/Ctrl + S** after each step is good muscle memory. It also forces a quick checkpoint so if anything breaks, you can revert to the last saved state.

05 Add the Gmail trigger

The trigger is what kicks off the workflow. We want it to run whenever a new email arrives in our Gmail inbox.

1. Click the "Add first step..." placeholder in the middle of the canvas
2. A search panel slides in from the right. In the search box, type
3. You'll see two results — pick **Gmail Trigger** (not "Gmail" — that one is for actions like *sending* mail; we'll add it later)
4. n8n will ask you to pick an event. Choose **Message Received**
5. You'll see the node appear on the canvas with a red badge — it needs configuration



Screenshot: Gmail Trigger node added, awaiting configuration

Drop a screenshot of the Gmail Trigger node with the right-hand config panel open.

• TRIGGER VS ACTION — THE DIFFERENCE

A **trigger** node has a special lightning-bolt icon and always sits at the very start of a workflow. An **action** node does something — sends, fetches, writes, updates. Same Gmail brand, two very different roles. Get the right one or your workflow won't start.

06 Connect your Gmail account

Now we'll authorise n8n to read your Gmail using OAuth. n8n never sees your password — Google issues a token instead, and you can revoke it any time from your Google account settings.

1. In the right-hand panel, click the **Credential to connect with** dropdown
2. Click **+ Create New Credential**
3. Pick **Sign in with Google** — a new browser tab opens
4. Choose the Google account you want to monitor
5. Google shows a permissions screen — review what n8n is asking for (read mail, manage drafts) and click **Allow**
6. You'll be returned to n8n. The credential dropdown now shows your Gmail account with a green check.

CONFIGURE THE TRIGGER PARAMETERS

Poll Times	Every Minute (for testing). You can switch to "Every 5 Minutes" once it's running.
Simplify Output	On (default) — gives you cleaner data fields like <code>subject</code> , <code>from</code> , <code>textPlain</code>
Filters	Leave empty for now. Later you can filter by label, sender, etc.

TEST IT

1. Click **Execute step** (the play button at the bottom-right of the panel)
2. n8n fetches your latest emails. You should see one or more results appear in the Output panel on the right.
3. Look for fields like `subject`, `from`, `textPlain`, and `threadId` — these are the bits we'll use later.

• IF NO EMAILS SHOW UP

Send yourself a test email from another address (subject: *Test refund request*, body: a sentence or two). Then click **Execute step** again. The trigger only returns recent unread mail; older mail won't appear.

07 Classify the email with AI

Now the interesting part. We'll send the email to OpenAI and ask it to figure out three things: what category the query falls into, how urgent it is, and a one-line summary. We tell the AI to give us a structured JSON response so the rest of our workflow can use the data cleanly.

1. Click the small **+** on the right edge of your Gmail Trigger node
2. In the search panel, type `OpenAI`
3. Click **OpenAI** (the icon with the green leaf)
4. For Resource, choose **Message a Model**

CONNECT OPENAI

5. Click the credentials dropdown → **+ Create New Credential**
6. Paste the OpenAI API key you saved in Step 2 (starts with `sk-`)
7. Click **Save**. The credential is now stored, encrypted, in your n8n account.

SET THE PARAMETERS

Model	<code>gpt-4o-mini</code> — fast, cheap, more than smart enough for this task
Output Format	JSON (this forces a clean response — no markdown, no extra text)
Simplify Output	Leave on

THE SYSTEM MESSAGE (THE AI'S JOB DESCRIPTION)

Click **Add Message** → choose role **System**, then paste this prompt:

SYSTEM MESSAGE

You are a customer support classifier. Read the email below and respond with ONLY a valid JSON object in this exact format:

```
{
  "category": "refund | technical | billing | general | complaint",
  "priority": "low | medium | high | urgent",
  "summary": "one sentence describing the customer's issue"
}
```

Choose exactly ONE value for category and ONE for priority.
Return only the JSON. No markdown, no code fences, no preamble.

THE USER MESSAGE (THE EMAIL ITSELF)

Add another message → role **User**. This time, instead of typing, drag fields in from the left-hand input panel. The `{{ }}` syntax is how n8n references data from earlier nodes.

USER MESSAGE

Subject: `{{ $json.subject }}`

From: `{{ $json.from }}`

Body:
`{{ $json.textPlain }}`

• HOW TO DRAG, NOT TYPE

In the left panel of the OpenAI node, you'll see the data from your Gmail Trigger — fields like `subject`, `from`, `textPlain`. Click the field name and drag it into the message body. n8n writes the `{{ $json.fieldname }}` expression for you. This is faster and avoids typos.

TEST IT

1. Click **Execute step**

2. In the Output panel, you should see a JSON object with three fields: `category`, `priority`, `summary`

3. If you see a string instead of a structured object, check that **Output Format** is set to JSON



Screenshot: OpenAI classification result with JSON output

Drop a screenshot of the OpenAI node showing a successful JSON classification.

- **A TRICK FOR PROMPT RELIABILITY**

The line *"No markdown, no code fences, no preamble"* is doing real work here. Without it, models sometimes wrap JSON in ````json` blocks, which then breaks downstream parsing. Always be explicit about format expectations in your prompts.

08 Tidy up the data with an Edit Fields node

Right now we have data scattered across two earlier nodes — the email lives in the Gmail Trigger output, the classification lives in the OpenAI output. The next steps will need both, and pulling from two places gets messy. Let's combine everything into a single clean set of fields.

1. Click the + on the right edge of your OpenAI node
2. Search for `Edit Fields` and click it (sometimes called "Set" in older versions)
3. For Mode, choose **Manual Mapping**
4. Click + **Add Field** for each of the values below

THE FIELDS TO SET

emailFrom	<code>{{ \$('Gmail Trigger').item.json.from }}</code>
emailSubject	<code>{{ \$('Gmail Trigger').item.json.subject }}</code>
emailBody	<code>{{ \$('Gmail Trigger').item.json.textPlain }}</code>
threadId	<code>{{ \$('Gmail Trigger').item.json.threadId }}</code>
category	<code>{{ \$json.message.content.category }}</code>
priority	<code>{{ \$json.message.content.priority }}</code>
summary	<code>{{ \$json.message.content.summary }}</code>
timestamp	<code>{{ \$now.toISO() }}</code>

5. Set **Include Other Input Fields** to **OFF** — we want a clean slate
6. Click **Execute step**
7. In the output, confirm all eight fields are populated

• WHAT THESE EXPRESSIONS MEAN

`$('Gmail Trigger').item.json.subject` reaches back to the Gmail Trigger node and grabs the `subject` field. `$json` on its own refers to the *previous* node (the OpenAI one). `$now.toISO()` is a built-in helper that gives you the current timestamp as a clean ISO string. You'll use these patterns constantly.

09 Have AI draft the reply

One more OpenAI call. This time we want a friendly, professional reply — not JSON. We give the AI the original email plus the classification we just made, and ask for plain text.

1. Click the + after your Edit Fields node
2. Search `OpenAI` again, choose the same OpenAI node
3. Resource: **Message a Model**
4. Credentials: pick the OpenAI credential you already saved
5. Model: `gpt-4o-mini`
6. Output Format: **Text** (not JSON this time — we want a real reply)

THE SYSTEM PROMPT

Add a System message:

SYSTEM MESSAGE

You are a friendly, professional customer support agent for [Your Company]. Write a warm reply to the customer based on their email and the classified summary.

Rules:

- Keep it under 150 words
- Acknowledge the issue clearly in the first line
- Don't make commitments you can't keep — no specific refund timelines or compensation amounts
- Sign off with: "Best regards, The Support Team"
- If priority is "urgent" or category is "complaint", add this line near the end: "I'm escalating this to a senior team member who will follow up within 24 hours."

Output only the email body. No subject line, no greeting metadata, no markdown.

Replace `[Your Company]` with your actual company or product name.

THE USER MESSAGE

Add a User message that hands the AI both the original email and the classification:

USER MESSAGE

Customer email subject: {{ \$json.emailSubject }}
From: {{ \$json.emailFrom }}
Issue category: {{ \$json.category }}
Priority: {{ \$json.priority }}
One-line summary: {{ \$json.summary }}

Original email body:
{{ \$json.emailBody }}

Write the reply now.

TEST IT

1. Click **Execute step**
2. Read the output. Is it warm? Professional? On-brand?
3. If something's off — too long, too cold, wrong tone — adjust the system prompt and re-run. This is the iterative part of working with AI.

• THE "NO COMMITMENTS" RULE, EXPLAINED

AI models can confidently promise things like *"We'll refund you within 24 hours"* when they shouldn't. That promise becomes a customer expectation, and breaking it damages trust. The system prompt explicitly forbids specific commitments — the AI acknowledges and escalates instead. This is the kind of guardrail that separates safe automations from dangerous ones.

• TRY A FEW DIFFERENT TEST EMAILS

Send yourself a refund request, a billing question, and an angry complaint. Run the workflow on each. Read the drafts. Tune the prompt until you'd be happy for these to go out from your real support team.

10 Save the reply as a Gmail draft

This is where we choose *not* to fully automate. Instead of sending the reply, we save it as a draft. A human at your support team opens Gmail, reviews the draft, edits if needed, and clicks send. Five seconds of human attention prevents 99% of AI mishaps.

1. Click the **+** after your second OpenAI node
2. Search `Gmail` and pick **Gmail** (the action node, not the trigger)
3. Resource: **Draft**
4. Operation: **Create**
5. Credential: pick the same Gmail credential you connected in Step 6

MAP THE FIELDS

Subject	Re: {{ \$('Edit Fields').item.json.emailSubject }}
Email Type	Text
Message	{{ \$json.message.content }}
Send To	{{ \$('Edit Fields').item.json.emailFrom }}

ADD OPTIONS → THREAD ID

1. Click **Add Option**
2. Pick **Thread ID**
3. Set it to: {{ \$('Edit Fields').item.json.threadId }}

This keeps the draft inside the same email conversation, so when your support team opens it, it appears nested under the original customer email — exactly where they expect to see it.

TEST IT

1. Click **Execute step**
2. Open Gmail in another tab, click **Drafts**
3. You should see a new draft with the reply text, threaded under the original email

- **WHY "DRAFT" IS THE SAFER DEFAULT**

Auto-sending replies feels magical when it works and catastrophic when it doesn't. Drafts give you the speed gain (90% of the work is done) without the risk (a human still hits send). Once you've tested the workflow for a few weeks and the drafts are consistently good, *then* you can graduate categories to auto-send one at a time.

11 Log every query to Google Sheets

Last node. We append a row to the Customer Support Log sheet you created in Step 3, so your team has a running record of every email handled.

1. Click the + after your Gmail Draft node
2. Search `Google Sheets` and pick the **Google Sheets** node
3. Resource: **Sheet** (or "Sheet within Document" depending on n8n version)
4. Operation: **Append Row**
5. Credential: + **Create New Credential** → Sign in with Google → pick the same Google account → allow permissions
6. Document: pick **Customer Support Log** from the dropdown (n8n lists your sheets)
7. Sheet: pick **Sheet1**
8. For Mapping Column Mode, choose **Map Each Column Manually**

MAP THE COLUMNS

Timestamp	<code>{{ \$('Edit Fields').item.json.timestamp }}</code>
From	<code>{{ \$('Edit Fields').item.json.emailFrom }}</code>
Subject	<code>{{ \$('Edit Fields').item.json.emailSubject }}</code>
Category	<code>{{ \$('Edit Fields').item.json.category }}</code>
Priority	<code>{{ \$('Edit Fields').item.json.priority }}</code>
Summary	<code>{{ \$('Edit Fields').item.json.summary }}</code>
Draft Status	Draft created

TEST IT

1. Click **Execute step**
2. Open your Google Sheet — a new row should appear with all fields populated

• IF A COLUMN SAYS "UNDEFINED"

Check that your sheet's column headers exactly match the names in the table above (case sensitive). Also confirm Edit Fields ran successfully — if the upstream node has empty data, sheet cells will show empty or undefined.

12 Run the whole workflow end-to-end

You've tested each node in isolation. Now let's run the full chain — one email in, one draft and one logged row out.

1. Save your workflow (**Cmd/Ctrl + S**)
2. Send yourself a test email from another address. Make it realistic — try one of these subjects:
 - "I want a refund for my order #4582"
 - "Cannot reset my password — locked out for 2 days"
 - "Charged twice for my subscription this month"
3. Wait up to 60 seconds for the Gmail Trigger to poll (you set it to "Every Minute" in Step 6)
4. Click **Executions** at the top of the page
5. You should see a new execution. Click into it.

WHAT "SUCCESS" LOOKS LIKE

- All five nodes show a green check
- The Gmail Trigger output contains your test email
- The first OpenAI node returns a clean JSON object with category, priority, summary
- The Edit Fields node has all eight fields populated
- The second OpenAI node returns a draft reply that's warm and concise
- Gmail's Drafts folder has a new draft, threaded under your test email
- Google Sheets has a new row with all the data



Screenshot: Successful execution timeline showing all green nodes

Drop a screenshot of the executions view with all-green nodes here.

• IF SOMETHING FAILS

n8n will mark the failing node in red. Click into it to see the exact error. The most common causes are missing credentials, a typo in an expression, or hitting an API rate limit. Part 5 of this workbook covers the top issues — it's the next stop if you're stuck.

13 Activate it in production

So far, your workflow only runs when you click "Execute." Activating flips it to live mode — the Gmail Trigger now polls automatically on the schedule you set, and every new email triggers the chain without you doing a thing.

1. In the top bar, find the **Inactive** toggle on the right
2. Click it — the toggle slides to **Active** with the colour **green**
3. n8n shows a confirmation: *"Workflow is now active"*
4. Save once more to be safe (**Cmd/Ctrl + S**)

That's it. From now on, every email that lands in your inbox will trigger this workflow within 60 seconds. You can close the n8n tab — it runs in the cloud regardless of whether your browser is open.

TWEAKING AFTER LAUNCH

You'll probably want to adjust two things in the first week of running this:

- **Poll frequency.** Every minute is great for testing but uses your free executions quickly. Drop it to "Every 5 Minutes" once you're confident it works.
- **Email filters.** Right now the workflow runs on *every* email — including newsletters, internal mail, and personal stuff. Add a filter in the Gmail Trigger to only fire on emails sent to `support@yourdomain.com`, or a specific Gmail label.

• WATCH THE EXECUTIONS TAB FOR A FEW DAYS

The first 48 hours after activation are when you'll catch edge cases — emails the AI misclassifies, weird formatting, replies that miss the tone. Open the Executions view daily, sample a few runs, and tune the prompts. After a week the workflow will feel like it's been running forever.

You shipped it. ✨

A working AI customer support workflow, built from scratch, running on your account. This is the same shape as the automations real teams use in production. Take a moment.

PART FIVE

Troubleshooting

Every problem you'll hit is in the next two pages. Find the symptom, follow the fix.

When things don't work

Workflows fail. That's normal. n8n is unusually good at telling you *why*. Click the failing node, scroll to the error message, and match it against the list below.

OpenAI errors

"INSUFFICIENT QUOTA" OR "BILLING ISSUE"

You haven't added a payment method to OpenAI, or your prepaid balance is at \$0. Open `platform.openai.com` → Billing → Add credits. The error will clear within a minute.

"INVALID API KEY"

Your key was rotated or you copied it wrong. Generate a new one at `platform.openai.com/api-keys`, update the credential in n8n (Credentials menu, top-left).

OPENAI RETURNS TEXT INSTEAD OF JSON

Usually the model wrapped the JSON in a ````json` markdown block. Two fixes: confirm **Output Format** is set to **JSON** on the node, and add this to your system prompt: *"Return raw JSON only. No markdown, no code fences."*

RATE LIMIT ERRORS

You hit OpenAI's per-minute limit. For a beginner workflow this almost never happens — but if it does, switch your trigger from "Every Minute" to "Every 5 Minutes" and add a Wait node between OpenAI calls if needed.

Gmail errors

TRIGGER NEVER FIRES

Three things to check, in order:

1. Is the workflow toggle set to **Active**?
2. Did you wait long enough? (Trigger polls every 1–5 minutes — be patient)
3. Did your test email actually arrive in your Gmail inbox? Check Spam.

"INSUFFICIENT PERMISSIONS"

The Gmail OAuth token expired or you revoked it. In n8n, go to Credentials → Gmail → Reconnect.

DRAFT CREATED BUT NO THREAD / WRONG THREAD

You probably mapped the wrong field for Thread ID. It should be `{{ $('Edit Fields').item.json.threadId }}`. If empty, the draft becomes a new conversation rather than a reply.

Google Sheets errors

"CANNOT FIND SPREADSHEET"

You probably picked a sheet from a different Google account. Make sure the credential you connected for Sheets is the same Google account that owns the Customer Support Log sheet.

CELLS SAY "UNDEFINED" OR ARE EMPTY

The data isn't reaching that column. Click the previous Edit Fields node, run it manually, and confirm all eight fields are populated. Also double-check that your sheet's column headers match exactly (case sensitive).

"RANGE NOT FOUND"

You probably renamed your sheet tab from "Sheet1" to something else. Either rename it back to `Sheet1`, or update the Sheet field on the Google Sheets node to match.

n8n itself, and general fixes

EXPRESSIONS SHOW AS RED TEXT

Red means the expression is invalid — usually a typo or a field that doesn't exist on that node yet. Click into the field, look at the expression carefully, and try dragging the value from the left panel instead of typing.

WORKFLOW RUNS BUT NO DATA FLOWS DOWNSTREAM

One of your nodes returned an empty result. Check each node's output panel from left to right. The first one with empty output is your problem. Often it's the OpenAI node hitting an unusual input — check the input panel to see what reached it.

FREE TRIAL RAN OUT

The n8n Cloud free trial is 14 days. After that, you have two options:

1. Upgrade to a paid plan (starts around \$24/month for 2,500 executions)
2. Self-host n8n with Docker — it's free, takes about 10 minutes to set up, and runs the same workflows you've already built. Search "n8n self-host Docker" on the n8n docs site.

"MY WORKFLOW WORKED YESTERDAY AND NOW IT DOESN'T"

This usually means a credential expired (Gmail OAuth tokens last 7 days for some account types, longer for others). Go to Credentials, find the one with a yellow warning, click Reconnect.

THE AI'S CLASSIFICATIONS FEEL OFF

Not really an error — but worth saying. If the model keeps mislabelling things, the system prompt usually needs more explicit definitions. Add a few examples, or list edge cases:

A "complaint" specifically means the customer is expressing anger or frustration. A polite refund request is "refund", not "complaint", even if it mentions a problem.

Iterate on the prompt three or four times. The difference between an okay automation and a great one is almost always prompt quality.

• STILL STUCK?

Two best resources, in order: **1.** The n8n community forum at community.n8n.io — searchable, friendly, fast responses. **2.** Click the "?" icon in n8n itself for context-sensitive docs on whichever node is open.

PART SIX

Stretch challenges

*Four ways to make your workflow smarter, safer, or more useful.
Pick whichever sounds most fun.*

Four challenges to take it further

If the basic workflow felt easy, here are four ways to extend it. Each one introduces a new n8n concept worth knowing.

A Auto-send for low-risk categories

Right now everything goes to drafts. But for low-priority general inquiries, drafts might be overkill — a simple "We've received your question, expect a reply within 24 hours" is fine to auto-send.

What to add: An **IF** node after Edit Fields. The condition: `category equals "general"` AND `priority equals "low"`. The "true" branch goes straight to **Gmail → Send Message**. The "false" branch goes through the existing draft flow.

What you'll learn: Branching logic with IF nodes, the most-used flow control feature in n8n.

B Slack alert for urgent emails

When a complaint or urgent issue comes in, your support team probably isn't watching Gmail in real time. Ping them on Slack instead.

What to add: An IF node checking if `priority equals "urgent"` OR `category equals "complaint"`. On the true branch, add a **Slack → Send Message** node. Map the message body to include the email summary and a link to the Gmail draft.

What you'll learn: Multi-output workflows and Slack integration.

C Knowledge base lookup before drafting

If you have a Notion page or Google Doc with FAQs, give the AI access to it. Before drafting a reply, it can pull the most relevant FAQ and incorporate it.

What to add: Before the second OpenAI node, add a **Google Docs** or **Notion → Get Page** node to pull your FAQ content. Then pass that content into the AI's user message as additional context: *"Reference this knowledge base when relevant: {{ \$json.faqContent }}"*.

What you'll learn: The basics of retrieval-augmented generation (RAG) — giving an AI access to your own data.

D Daily summary email of all queries

Build a second workflow that runs every morning, reads yesterday's rows from your Customer Support Log, and emails you a summary: how many queries, breakdown by category, top 3 most urgent, etc.

What to add: A new workflow with a **Schedule Trigger** set to 9am daily. Then **Google Sheets** → **Read Rows** with a filter for yesterday's date. Pipe that into an **OpenAI** node asking for a written summary. Finally a **Gmail** → **Send Message** to your own inbox.

What you'll learn: Schedule triggers (the most common pattern after webhook triggers) and the read-and-summarise pattern, which works for hundreds of use cases.

A general principle for everything you build next

Every automation you'll ever build looks roughly like this: **trigger** → **transform** → **decide** → **act**.

Something kicks it off, you reshape the data, you make a decision (often with AI now), then you take an action. Once you see the pattern, every new tool — Slack, Notion, Stripe, anything — slots into one of those four roles.

- **WHERE TO FIND MORE INSPIRATION**

Visit n8n.io/workflows for thousands of community-built templates you can import and adapt. Looking at a real workflow built by someone else and reverse-engineering it is one of the fastest ways to learn.

PART SEVEN

Glossary

The vocabulary that comes up over and over. Bookmark this page; you'll reference it for weeks.

The terms that come up most

Workflow

A connected sequence of nodes that runs together. Triggered by an event or schedule, runs once, ends.

Node

A single step in a workflow. Each node either fetches data, transforms it, makes a decision, or performs an action on an external service.

Trigger

A special node that starts a workflow. Common types: webhook (incoming HTTP), schedule (timed), or app-event (like "new email" or "new row in sheet").

Action node

Any non-trigger node that does something — sends, fetches, updates, transforms. Most of your nodes will be action nodes.

Execution

One run of a workflow, from trigger to last node. n8n logs every execution, including failures, with full input/output data so you can debug after the fact.

Credential

Stored authentication for an external service. Set up once, reusable across nodes. Encrypted at rest.

Expression

A small piece of code wrapped in `{{ }}` that pulls data from earlier nodes or uses built-in helpers like `$now`. Expressions are how data flows through your workflow.

JSON

A simple text format for structured data. The lingua franca of APIs. n8n passes JSON between nodes by default.

OAuth

An authentication method where you grant a service permission to act on your behalf without sharing your password. Used for Gmail, Google Sheets, Slack, and most modern apps.

Polling

The technique a trigger uses to check for new data on a schedule (e.g. every 5 minutes). Less efficient than webhooks but works with any service.

Webhook

A URL that an external service can POST data to. The instant a new event happens, the workflow fires — no polling delay. Faster than polling, but the external service has to support it.

Prompt

The instructions you give to an AI model. Includes the system message (the AI's job description) and the user message (the specific request). Better prompts produce better outputs.

System message

The first instruction in a chat with an AI model. Sets the role, rules, and constraints. Stays consistent across the whole conversation.

Token

The unit of pricing for AI APIs. Roughly 4 characters of English text per token. A short customer email plus reply is ~500–800 tokens; gpt-4o-mini costs fractions of a cent for that.

Hallucination

When an AI confidently states something untrue. The reason your workflow saves drafts instead of auto-sending — a brief human review catches hallucinations before they reach customers.

Human-in-the-loop

The principle of keeping a person in control of the final action, even when AI does the heavy lifting. Drafts instead of sends. Suggestions instead of auto-actions. Safer and almost always faster than full automation.

PART EIGHT

What's next

You shipped your first AI workflow. Here's how to keep the momentum going.

Keep the momentum

Most people who attend a workshop never build the thing afterwards. You did. That's the hardest part out of the way. Here are five concrete things to do this week — pick one, do it tomorrow.

1. Adapt this workflow to your real inbox

You built it on a personal account. Now point it at your team's actual support inbox. Add a Gmail filter so it only fires on emails sent to `support@` or labelled *Support*. Watch the drafts for a week. Tune the prompt.

2. Build the reverse — for sales leads

The same shape works for sales: new email → AI scores the lead → AI drafts a personalised first reply → log to your CRM. Build it tonight; it's the same nodes in a slightly different order.

3. Explore n8n's template gallery

Visit `n8n.io/workflows` — thousands of community-built workflows. Find one that interests you, import it, and reverse-engineer how it works. This is the fastest way to learn new patterns.

4. Replace one repetitive task in your week

What do you do every Monday morning that takes 20 minutes? Compiling a report? Reading a feed? Sending a status update? That task is probably an n8n workflow waiting to be built.

5. Show someone else what you built

Teaching is the best way to make knowledge stick. Walk a teammate through your workflow this week. Their questions will reveal what you actually understand and what's still hand-wavy in your head.

A final note

Automation is a craft, not a product. The tools change every six months — n8n, OpenAI, the integrations, the models. The thinking doesn't. Knowing when to automate, where to keep humans in the loop, how to make a workflow you can actually maintain — that's what compounds.

You've shipped one workflow. Build ten more this year. By the eleventh, you'll have a feel for this stuff that very few people have. That's a real edge.

- **USEFUL LINKS**

n8n docs: docs.n8n.io

n8n templates: n8n.io/workflows

n8n community: community.n8n.io

OpenAI docs: platform.openai.com/docs